



Habib University

Digital Logic and Design (EE-172 / CS-130 / CE-222)

Fall 2025

Instructors: Muhammad Farhan, Aamir Hassan, Ahmed Ali Mustansir,

Areeba Aziz Rajput.

Assignment No: 4

Release Date: 30th March, 2026

Due by: April 5th, 2026 11:59 PM

Total points: 100

Purpose:

The purpose of this assignment is to help you practice the number representation; its conversion; and the arithmetic operations in different base formats. In addition, this assignment will aid you to understand the process of representing a signed number in different formats; parity bits; representing complements of a number; and performing BCD arithmetic.

Instructions:

This assignment should be done individually.

All questions should be answered in **black ink only**.

Scan your answer sheet and upload it on LMS before the due date.

Grading Criteria:

Your assignments will be checked by instructor/TA.

You can also be asked to give a viva where you will be judged whether you understood the question yourself or not. If you are unable to correctly answer the question you have attempted right, you may lose your marks. No excuses/justifications will be entertained.

Zero will be given if the assignment is found to be plagiarized.

Untidy work or improper submissions will result in a reduction of your points.

Late submission penalty:

1-day late submission – 20% deduction of the maximum allowable marks

2-days late submission – 40% deduction of the maximum allowable marks

No submission will be accepted after two days of the original deadline

a)

$F(t)$	H	U	$F(t+1)$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

b)

$F \backslash HU$	00	01	11	10
0		1	1	
1	1		1	

$$F(t+1) = F'U + HU + FH'U'$$

c)

$F(t)$	$F(t+1)$	H	U		H	U
0	0	X	0		X	0
0	1	X	1	α	X	1
1	0	U'	X		X	H'
1	1	U	X		X	H

d) $H = \overline{F+X}$ $U = \overline{F(t)+Y}$

$F(t)$	X	Y	$F(t+1)$	H	U
0	0	0	1	1	1
0	0	1	0	1	0
0	1	0	1	0	1
0	1	1	0	0	0
1	0	0	1	0	0
1	0	1	1	0	0
1	1	0	1	0	0
1	1	1	1	0	0

2a)

Previous State	Next State		Output	
	X=0	X=1	X=0	X=1
a	b	f	0	0
b	d	c	0	0
c	e	f	0	0
d	g	a	1	0 ✓
e	c	d	0	0
f	f	b	1	1
g	g	h	0	1
h	g	a	1	0 ✓

d=h

b)

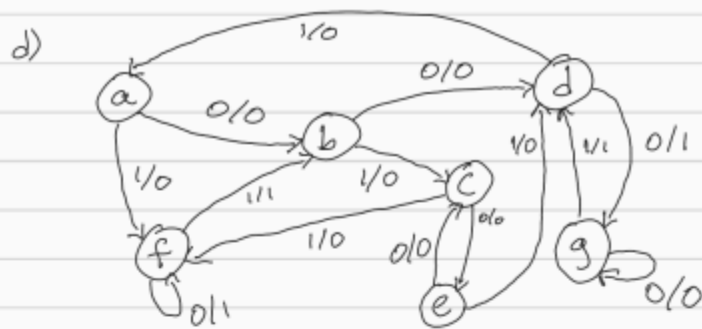
Previous State	Next State		Output	
	X=0	X=1	X=0	X=1
a	b	f	0	0
b	d	c	0	0
c	e	f	0	0
d	g	a	1	0
e	c	d	0	0
f	f	b	1	1
g	g	h	0	1

h eliminated because it is equivalent to d

c)

b	a b c					
c	b	a c				
d	X	X	X			
e	b f	a d	c e	X		
f	X	X	X	X	X	
g	X	X	X	X	X	X
h	X	X	X	a g	X	X
	a	b	c	d	e	f

hence d=h



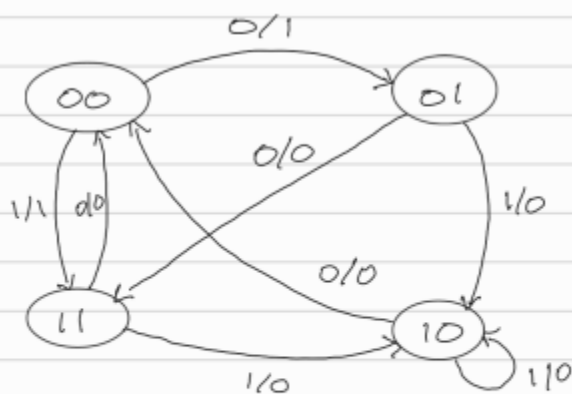
$$\begin{aligned}
 3a) \quad Q_1(t+1) &= J_1 Q_1' + K_1' Q_1 \\
 &= (x + Q_0) Q_1' + (x')' Q_1 \\
 &= x Q_1' + Q_0 Q_1' + x Q_1 \\
 &= x(Q_1 + Q_1') + Q_0 Q_1' \\
 &= x + Q_0 Q_1'
 \end{aligned}$$

$$\begin{aligned}
 Q_0(t+1) &= J_0 Q_0' + K_0' Q_0 \\
 &= Q_1' Q_0' + (Q_1 + x)' Q_0 \\
 &= Q_1' Q_0' + Q_1' x' Q_0
 \end{aligned}$$

b)

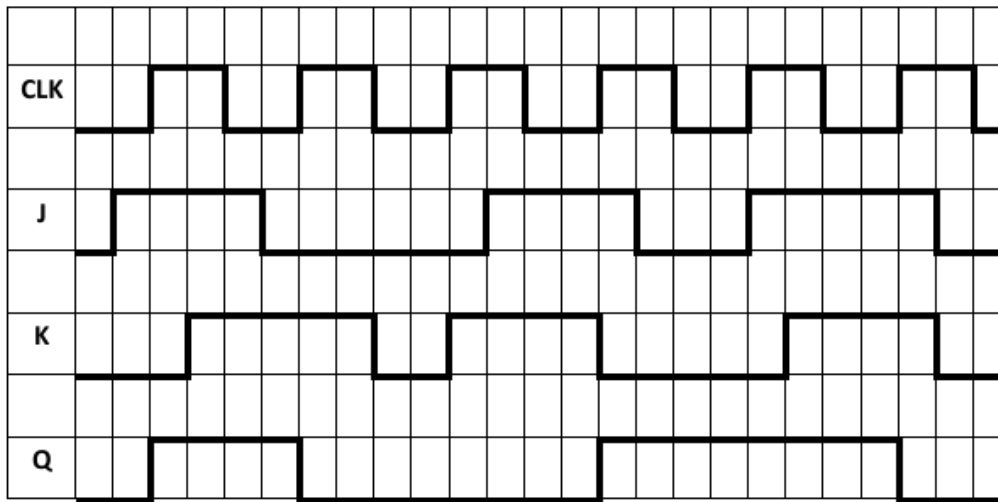
Current State		Input	Next State		JK Excitations				Output
Q_1	Q_0	x	Q_1^+	Q_0^+	J_1	K_1	J_0	K_0	y
0	0	0	0	1	0	1	1	0	1
0	0	1	1	1	1	0	1	1	1
0	1	0	1	1	1	1	1	0	0
0	1	1	1	0	1	0	1	1	0
1	0	0	0	0	0	1	0	1	0
1	0	1	1	0	1	0	0	1	0
1	1	0	0	0	1	1	0	1	0
1	1	1	1	0	1	0	0	1	0

c)

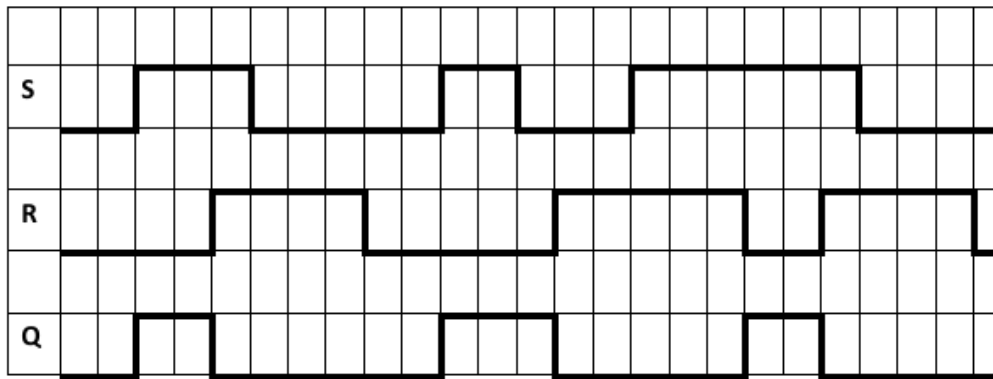


This is a Moore Machine because the output, $y = Q_1'Q_0'$ depends on current state only / does not depend on input x .

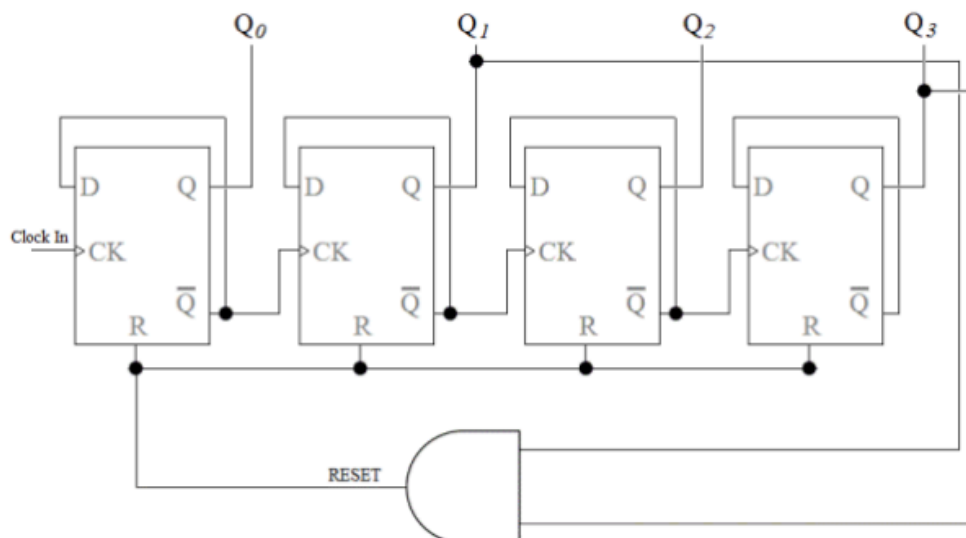
- a. Given the timing diagram of a positive edge triggered JK flip-flop, draw the output Q in the following timing diagram.



- b. The inputs for an SR-Latch are given in the following diagram. Complete the diagram by drawing the output Q. (Assume that output Q would be 0 initially)

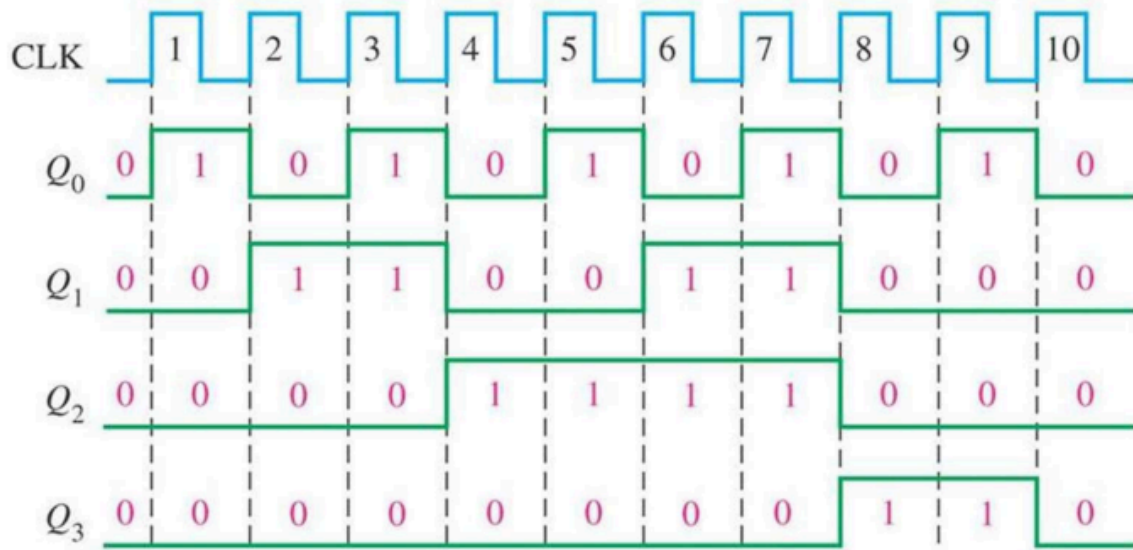


- a. For the given asynchronous circuit,



Draw its timing diagram and determine the function of the circuit.

Draw its timing diagram and determine the function of the circuit.

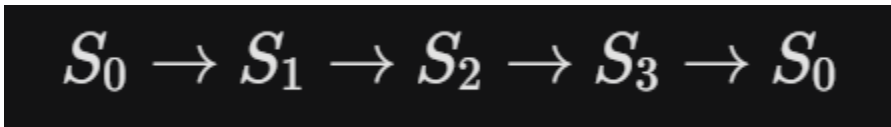


The circuit is an asynchronous BCD counter.

5

Design a sequential circuit using T flip-flops that simulates a digital clock module displaying the current time in 24 hour format. For example, if the current time is 23:12 then your circuit will display the binary value of 2, then 3, then 1 and then 2.

a. Draw the state diagram for the circuit above.



b. Draw the transition table for the circuit using T flip flops.

TIME: 11:40

Current State (Q_1Q_0)	Next State ($Q_1^+Q_0^+$)	Flip-Flop Inputs (T_1T_0)	Output ($Y_3Y_2Y_1Y_0$)
00 (S_0)	01 (S_1)	0 1	0 0 0 1
01 (S_1)	10 (S_2)	1 1	0 0 0 1
10 (S_2)	11 (S_3)	0 1	0 1 0 0
11 (S_3)	00 (S_0)	1 1	0 0 0 0

c. Use K-maps to deduce the equations for each T input for the flip flops.

4+4+2x6

		B	
		0	1
A	0	0	1
	1	2	3

$$T1 = Q0$$

		B	
		0	1
A	0	0	1
	1	2	3

$$F = 1$$

$$T0 = 1$$

		B	
		0	1
A	0	0	1
	1	2	3

$$F = AB'$$

$$Y2 = Q1Q0'$$

		B	
		0	1
A	0	0 1	1 1
	1	2	3

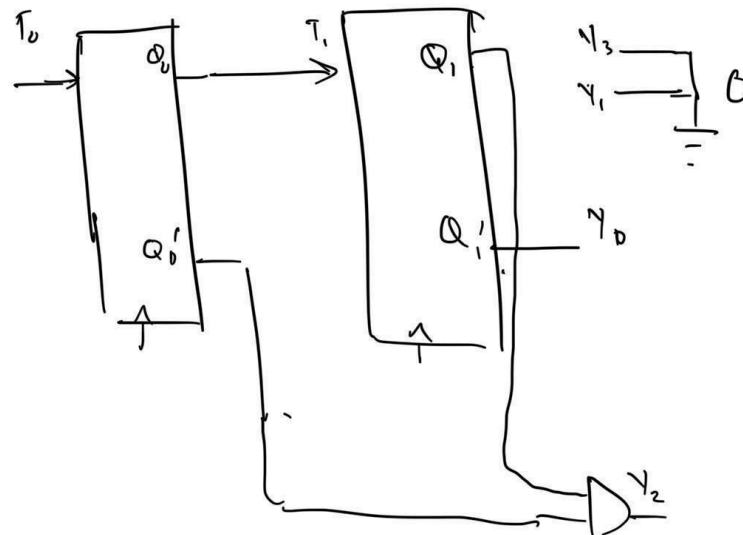
$$F = A'$$

$$Y1 = Q1'$$

Y3 Y1 would be all zeros so ground

So on so forth depending on values.

- d. Draw the circuit diagram for the sequential circuit.



Rough sketch, subjective based on time and approach to question

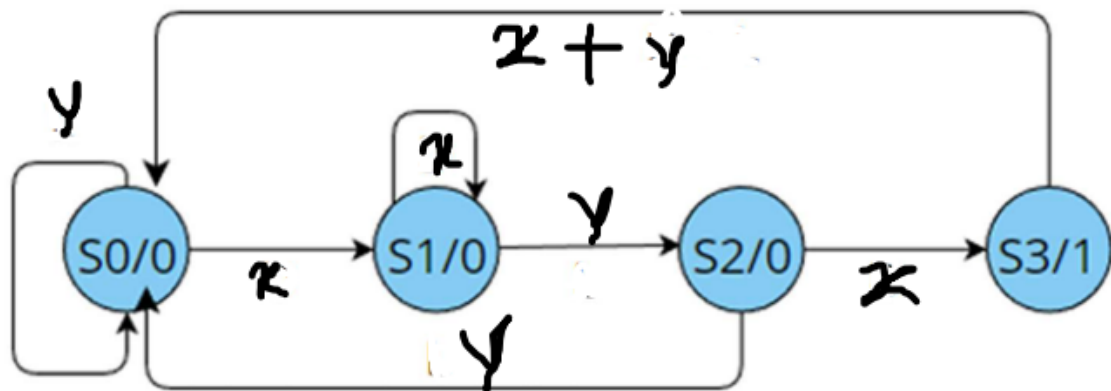
6

You are designing a smart door security system that unlocks only when a specific sequence of sensor signals is detected. Design a digital lock for the door using JK flip-flop which opens if a person inputs the sequence “XYX”. You can consider that you only have two inputs ‘X’ and ‘Y’ and one output Z (locked/ unlocked). Each element of the sequence must last for one or more cycles, and there may be zero or more cycles of both inputs low between the sequence elements. After the lock is open, either input going high causes the unlock to go low. Draw the circuit diagram for a digital lock implementation and show all the relevant working (state table, state diagram, K-maps, state equations). Specify what kind of state machine this is (mealy or moore)

15

Present State		Input		Next State		Output	FlipFlop A		FlipFlop B	
A	B	D	L	A	B	X	J _A	K _A	J _B	K _B
0	0	0	0	0	0	0	0	X	0	X
0	0	0	1	0	0	0	0	X	0	X
0	0	1	0	0	1	0	0	X	1	X
0	0	1	1	0	0	0	0	X	0	X
0	1	0	0	0	1	0	0	X	X	0
0	1	0	1	1	0	0	1	X	X	1
0	1	1	0	0	1	0	0	X	X	0
0	1	1	1	0	1	0	0	X	X	0
1	0	0	0	1	0	0	X	0	0	X
1	0	0	1	1	0	0	X	0	0	X
1	0	1	0	1	1	0	X	0	1	X
1	0	1	1	1	0	0	X	0	0	X
1	1	0	0	1	1	1	X	0	X	0
1	1	0	1	0	0	1	X	1	X	1
1	1	1	0	0	0	1	X	1	X	1
1	1	1	1	0	0	1	X	1	X	1

Replace DL with XY and output X with Z



Legend for States: S0 = 00, S1 = 01, S2 = 10, S3 = 11

Legend for States: S0 = 00, S1 = 01, S2 = 10, S3 = 11

Z <i>D,L</i> 00 01 11 10 <i>A,B</i> 00 0 0 0 0 01 0 0 0 0 11 1 1 1 1 10 0 0 0 0	Ja <i>D,L</i> 00 01 11 10 <i>A,B</i> 00 0 0 0 0 01 0 1 0 0 11 - - - - 10 - - - -
Jb <i>D,L</i> 00 01 11 10 <i>A,B</i> 00 0 0 0 1 01 - - - - 11 - - - - 10 - - - 1	Ka <i>D,L</i> 00 01 11 10 <i>A,B</i> 00 - - - - 01 - - - - 11 0 1 1 1 10 0 0 0 0
Kb <i>D,L</i> 00 01 11 10 <i>A,B</i> 00 - - - - 01 0 1 0 0 11 0 1 1 1 10 - - - -	

Replace DL with XY

State Equations for X, J_A, J_B, K_A and K_B:

$$X = AB$$

$$J_A = BD'L$$

$$J_B = D'L$$

$$K_A = BD + BL$$

$$K_B = D'L + AD$$

Circuit Diagram:

